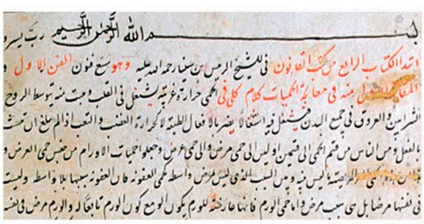
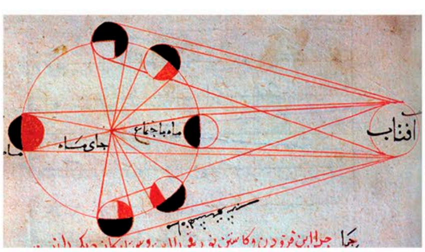
800–1300
ISLAMIC SCIENCE

The five centuries from 800 CE saw science flourish in the Islamic lands as scholars built on a legacy inherited from the classical empires and made advances of their own. The conquest of the eastern provinces of the Roman empire and the Persian Sasanian empire left the Islamic caliphate in control of important libraries and traditions of scholarship. The practical demands of medicine, architecture, and commerce, and the need to understand the geography of a vast empire led Muslim rulers to sponsor the translation of texts such as the Roman astronomer Ptolemy's *Almagest*.

The courts of the Umayyad and Abbasid caliphs at Damascus and Baghdad, and in regional centres such as Córdoba in Spain, became centres of learning, where experimentation and scientific advances outstripped those in Europe. Thinkers such as Al-Khwarizmi (c. 780–850 CE) developed algebra as a distinct discipline and provided solutions to quadratic equations, while in the 11th century, Ibn Sina (Avicenna) devised notions of impetus and inertia that went beyond the work of Aristotle. Al-Biruni (973–1048 CE) calculated the radius of Earth by measuring the height of a mountain.

From the 15th century, as new centralized Islamic empires emerged in India, Turkey, and Persia, Islamic science turned to producing compendia of existing knowledge rather than innovation. Its greatest age began to wane.





KEY MOMENTS

From c. 770 CE Translating classical works
Abbasid caliphs such as al-Mansur (r. 754–775 CE) established a library in Baghdad. This Bayt al-Hikma ("House of Wisdom") attracted scholars who translated classical manuscripts such as *De Materia Medica* by Dioscorides, (*left*) from Latin, Greek, and Persian into Arabic.

From c. 800 CE Islamic medicine
Physicians such as Persian Al-Razi (854–925 CE) made sophisticated observations (for example, distinguishing measles from smallpox for the first time), while the *Canon of Medicine* by Ibn Sina (980–1037) became a key medical textbook for centuries (*left*).

From c. 800 CE The physical sciences
Islamic scholars made great advances in optics, classified chemical substances for the first time, and produced sophisticated astronomical charts, such as this exposition (*left*) of the phases of the Moon produced by the scholar Abu Rayhan al-Biruni (973–1050).